Anouk Flinkert, KB National Library, The Hague and University of Bologna, 20-4-2026

Reconstructing Time in the Early Dutch Web:
Using Deep Neural Networks For Date Attribution

Digital Humanities Masther Thesis Internship

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The collection | XS4ALL

1992

First Dutch website www.nikhef.nl

1994-2001

Earliest homepages published on ISPs Euronet and XS4ALL

2007

KB starts collecting a selection of Dutch websites

2019

KPN announces discontinuation of hosting XS4ALL domain

KB 'rescues' ~3150 websites

2026

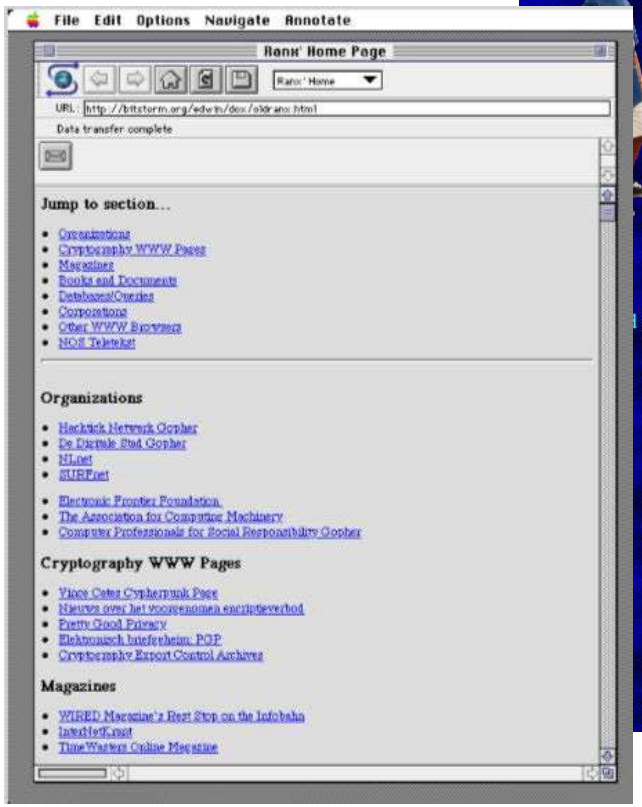
~25.000 websites

No domain crawl

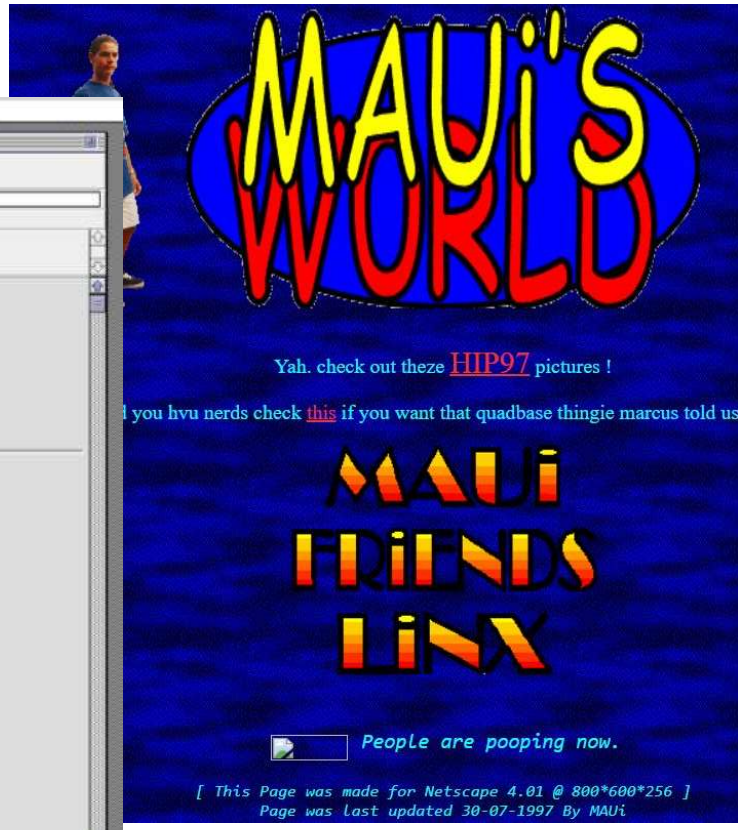
Welcome to the web of 2019!

Problem #1: Websites in SolrWayBack indexed with their harvest date (2019)

Luckily, metadata was documented during the acquisition of the collection.



Home page of user Ranx, juni 1994.



https://maui.home.xs4all.nl/ (1997) XS4ALL c

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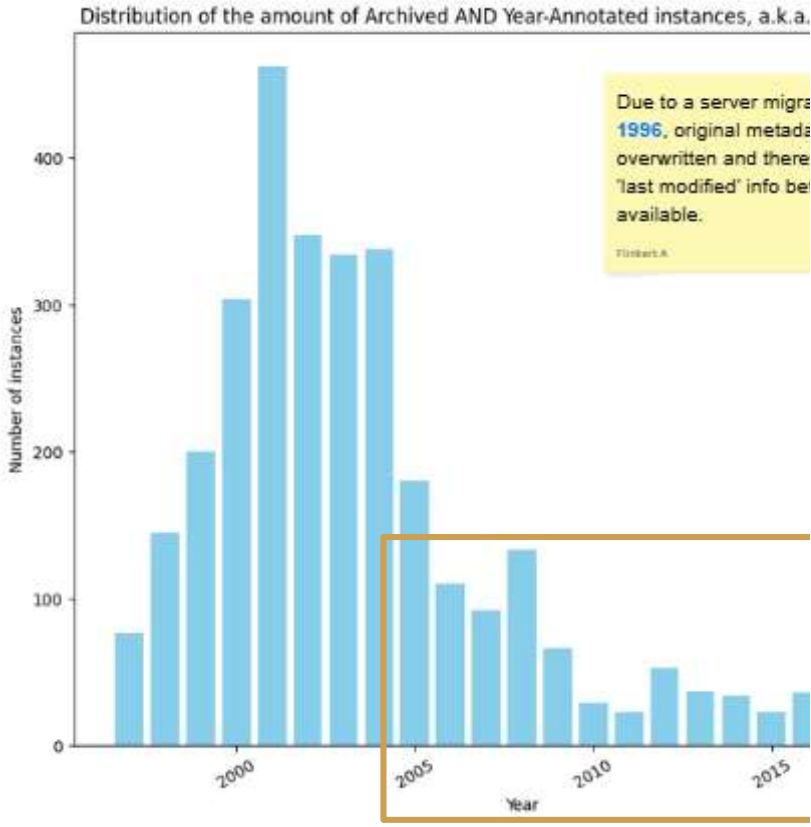
Metadata | XS4ALL 1997-2019?

Main method: *last updated* from object's 'page info'

Other close-reading methods for the ~1000 websites with an unknown last updated statement:

- Source-code traces
- Hidden metadata
- File sites
- Period specific software of terminology

Problem #2: Metadata does not reflect context (e.g. *'popular bike websites from 1999')*



Web philology | Looking under the hood

As in paleography and epigraphy, can we use script to date the texts?

Addressing a history and evolution of web development

From starting the HTML document instantly with <H1>...

```
<H1 ALIGN=CENTER><I>Welcome to Alain's HomePage</I></H1>

<P><I>There is no reason for any individual to have a computer in his home.<BR>
</I>- Kenneth H. Olson, President of DEC,Convention of the World Future
Society, 1977</P>
```

...and simplified JavaScript coding across browsers... and so on...

```
<script>
      <!-- Custom jQuery global variable to prevent conflict with page jQuery
      jqWayback = jQuery.noConflict(true);
</script>
```

```
          </td>
      </tr>
      </tbody></table>
</td>

</tr>
<tr>
```

..to using 50 tables to force more layout options...

Using Deep Neural Networks

Feature sets: raw source code extracted from the WARC, incl. html and inline CSS and JavaScr



Testing model performances of

RobertA

- Less computation time
- Better suited for fine-tuning
- Appropriate for many tasks

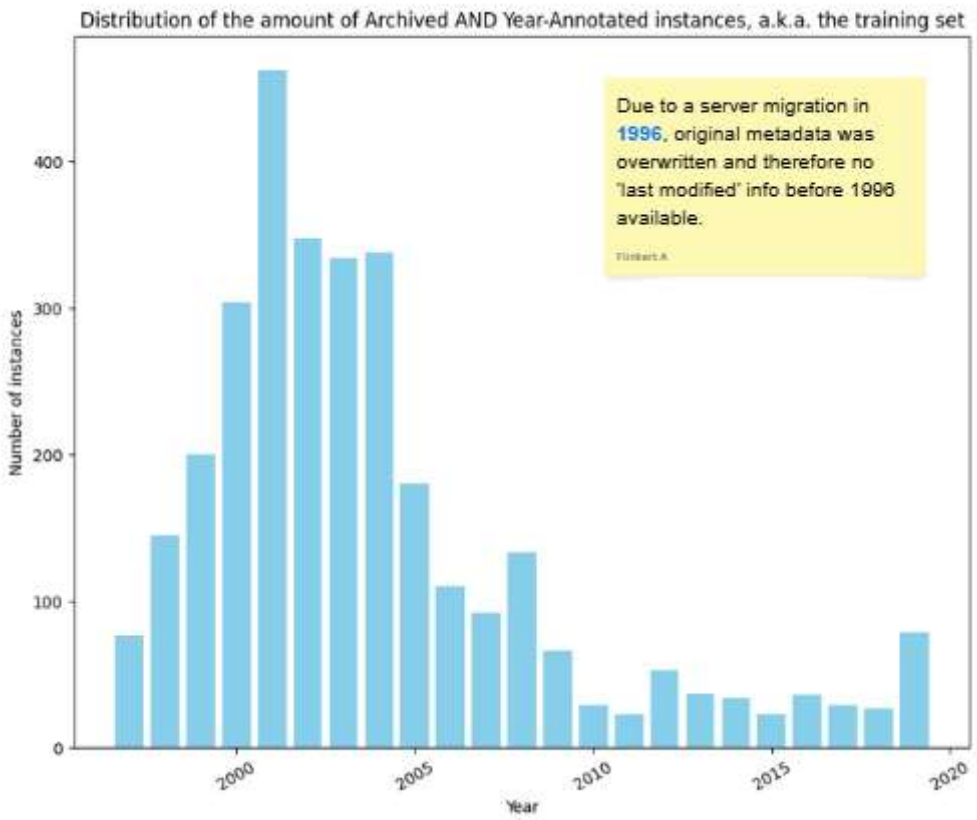
MarkupLM

- Maintains hierarchical relations between tags with XPath embeddings

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Using Deep Neural Networks | Ground truth

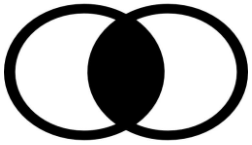
Current metadata based on 'last updated'



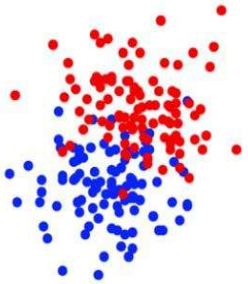
Issues we could find using this as a ground truth...



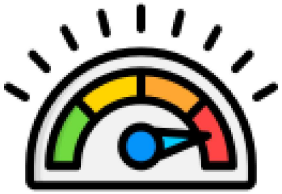
Class imbalance



Label noise



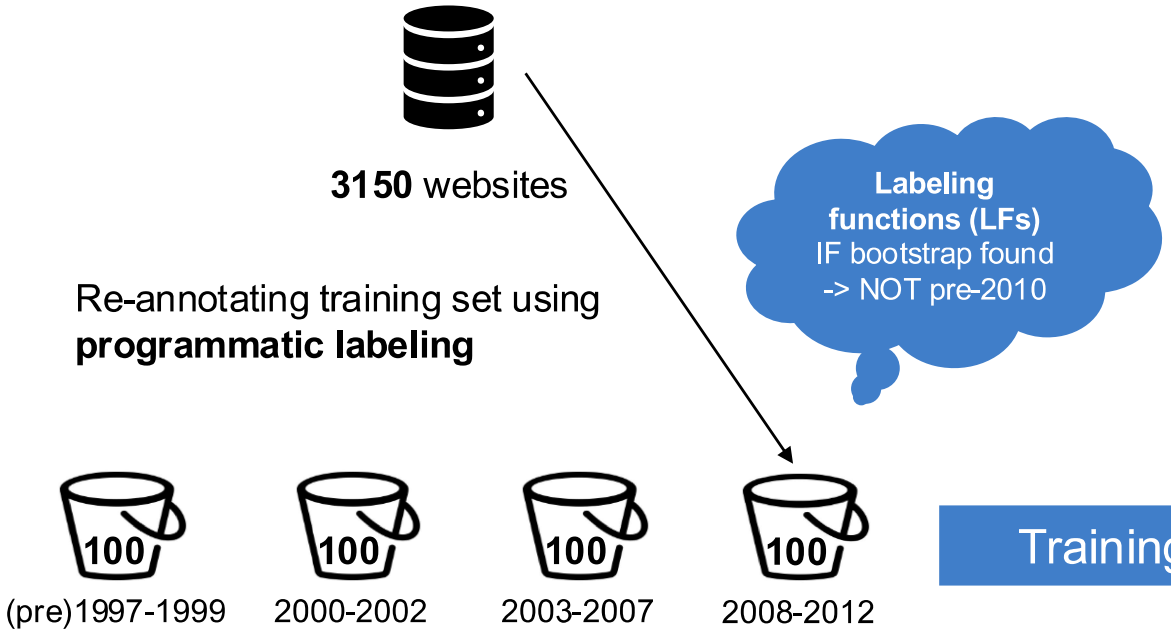
Intra-Class Variance



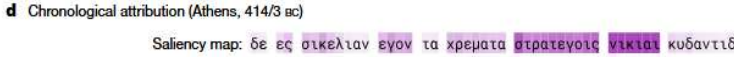
Token size

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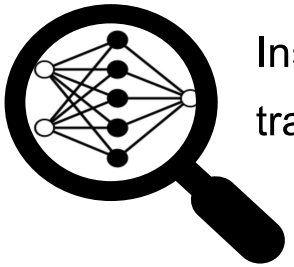
Using Deep Neural Networks | Current state



Example saliency map for chronological attribution



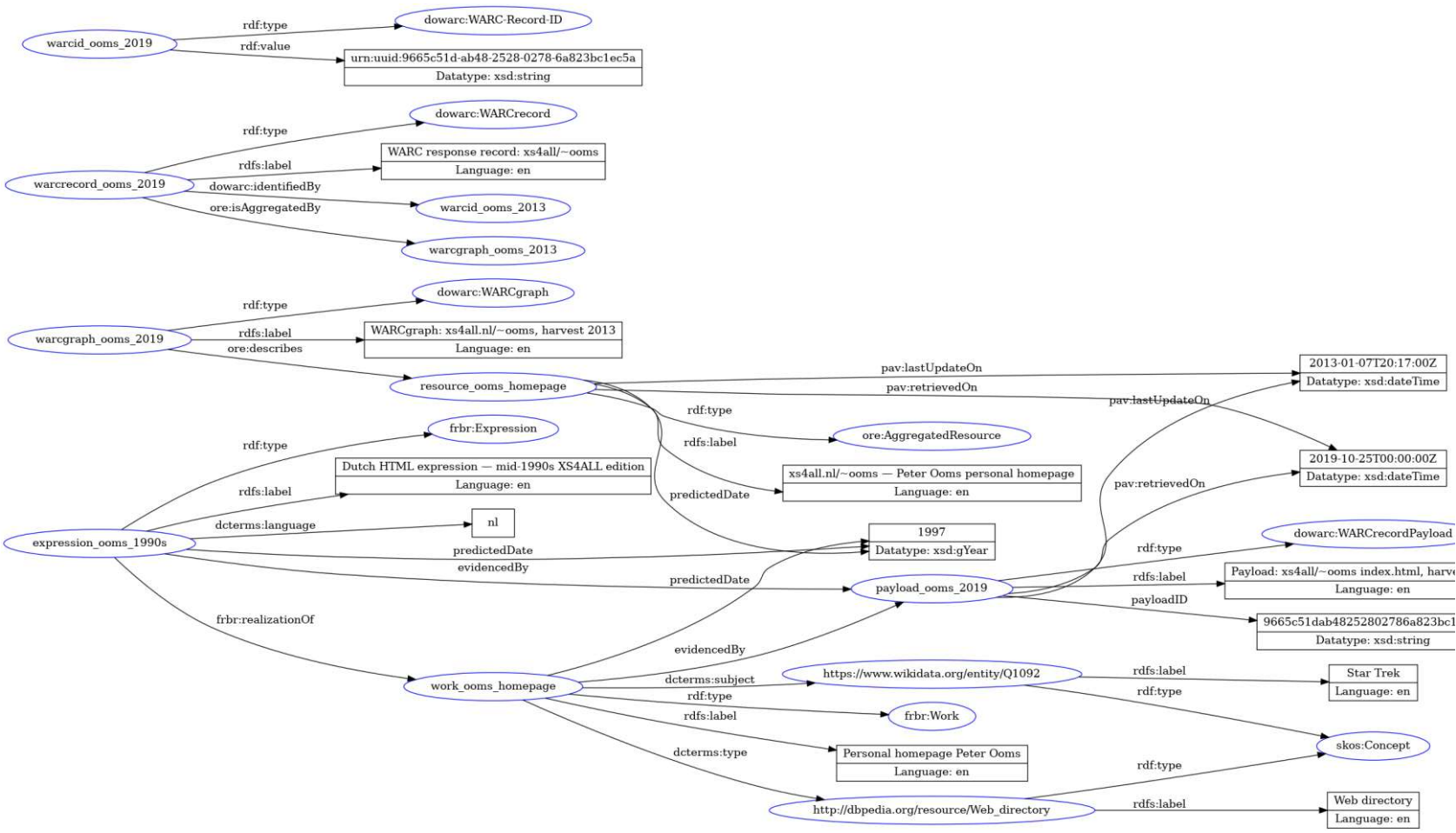
Assael, Y., Sommerschield, T., Shillingford, B. *et al.* Restoring and attributing using deep neural networks. *Nature* 603, 280–283 (2022). https://doi.org/10.1 022-04448-z



Inspecting the mode training using metrics

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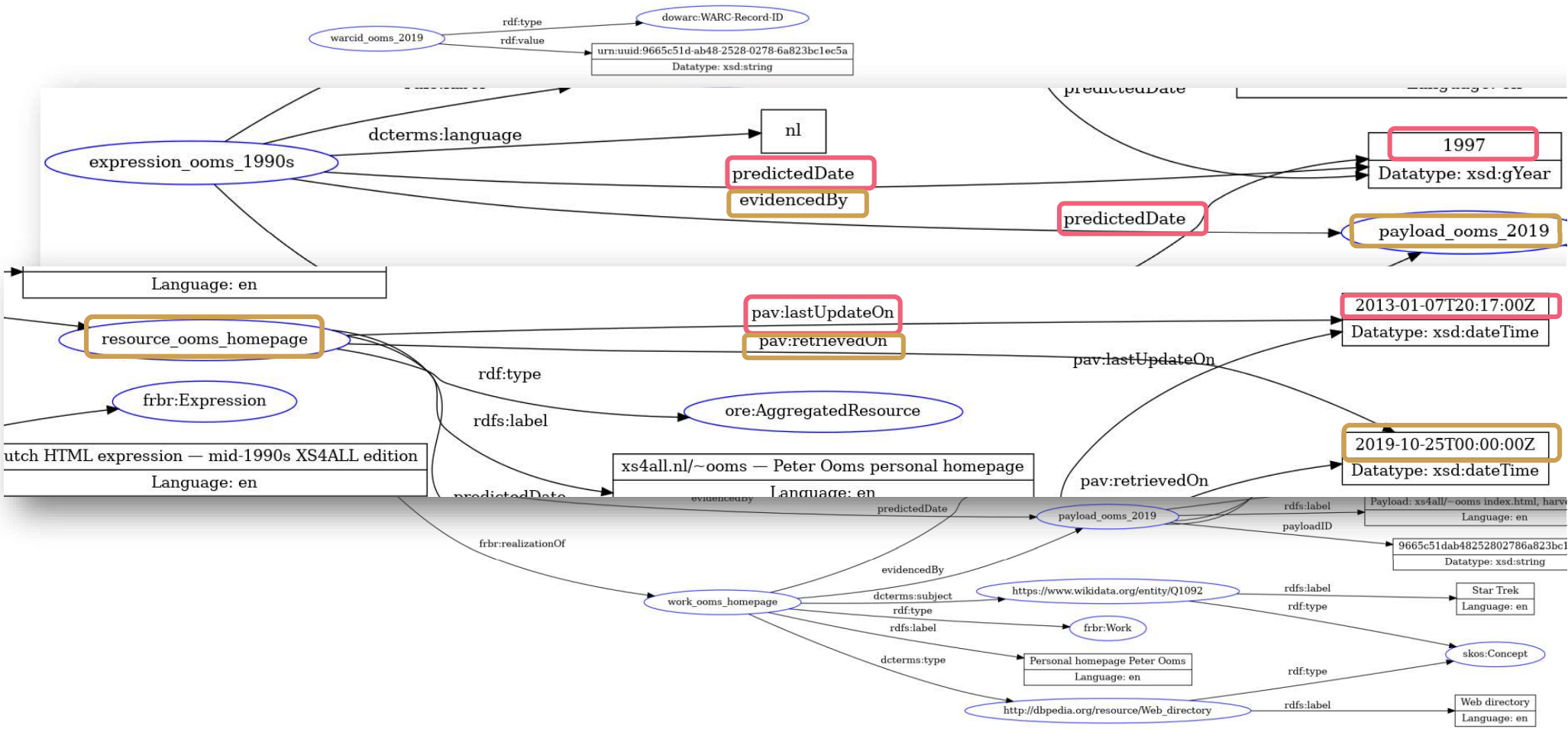
Knowledge Graphs | Storage and Access



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[1] *DOWARC*. Domain ontology of web archiving artefacts, such as WARC and CDX files, and web archiving activities. https://github.com/DOWARC/dowarc

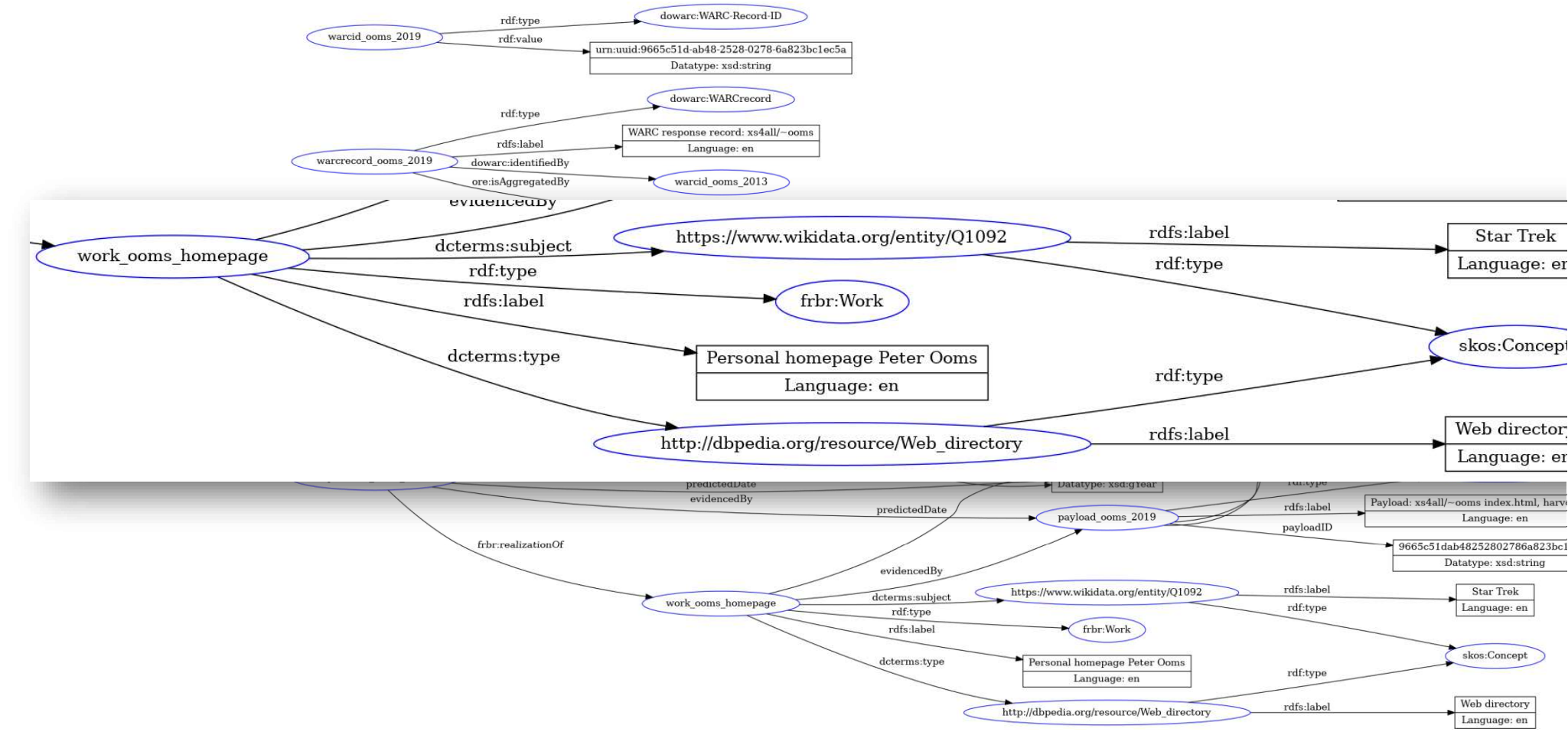
Knowledge Graphs | Managing multiple date



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[1] *DOWARC*. Domain ontology of web archiving artefacts, such as WARC and CDX files, and web archiving activities. https://github.com/DOWARC/dowarc

Knowledge Graphs | Semantic mapping



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AI in Archival Practice | Scope & Limitations

Technical scope

- 'Small' problem-specific AI solutions
- Era bucketing as a **first-pass filter**
- Predictions require **human validation** before annotation

Reliability & validity

- Small dataset
- **Collection bias**: curator preference for visually 'old' sites
- Scalability to modern, dynamic sites unclear
- Reproducibility: limited access to the archive

Scope of research

- Pattern- vs. outlier-driven analysis: which does this serve?
- What **hypothesis** does it answer?

These tools open new questions in web archival research, but do not yet resolve them.

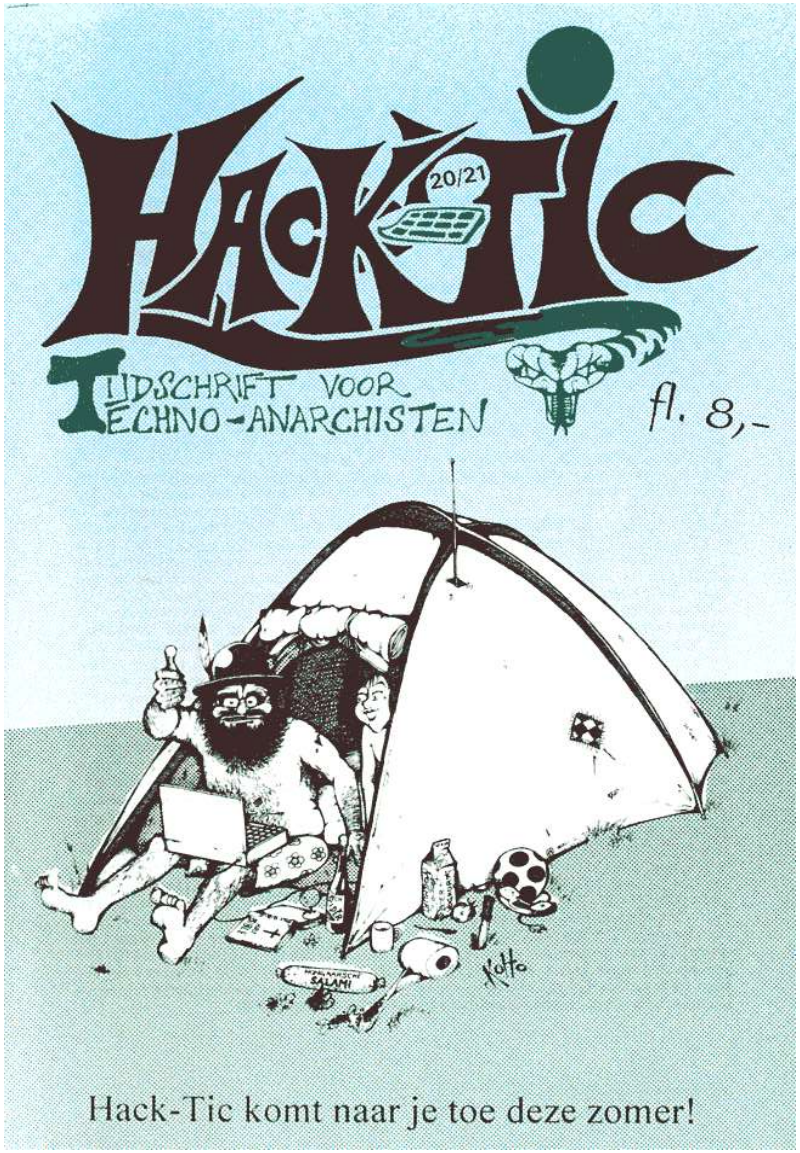
Thank you!

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https://github.com/digitalctrlv

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Magazine *Hack-Tic* edition 'Hacking at the End of the Universe' to announce the summer camp in 1994 in Lelystad by founders of Hack-Tic Netwerk, later XS4ALL. Illustrations by Koen Hottentot

Two research plans for

An **internship project** following applicability, the prototype and workflow within the organization



How do we make the XS4ALL collection searchable for users?



Application: accuracy, scalability and integration into the existing library system



A technical report, a functional prototype/pipeline and implementation advice

A **master thesis** contributing new theoretical knowledge to the field and specifically technical web phi



Why code as a historical source and do we interpret a DNN date attributio



Epistemology: the nature of the born digital object and the validity of the m



A 100 pages (~25.000-30.000 words) dissertation containing a scholarly argument that challenges or extends current academic theories

More specifically... | Research questions

Internship project

How can the integration of a deep learning classification pipeline and a structured knowledge base improve the archival workflow for the discovery, contextualisation, and retrieval of the XS4ALL web collection?

1. How reliably can a Deep Neural Network (DNN) identify and classify scripting features of websites within the XS4ALL collection for historical classification?
2. How can Explainable AI (XAI) and performance metrics provide insight into the model's decision-making process for these predictions?
3. *How can existing frameworks (such as BoDi, DOWARC, and FRBR) be extended to represent archived web objects within a knowledge base as both archival materials and cultural expressions?*
4. To what extent is the proposed pipeline scalable to the KB's full web archive, and what recommendations follow for its implementation?

Master thesis

How can deep neural networks (DNN) help to re
the early Dutch web within the XS4ALL collectio
what are the implications of these computational
methods for library and archival practice?

1. In the tradition of web philology, to what extent can
archived source code of *post festum* archived webs
as historical evidence for date attribution?
2. Using qualitative analysis of the model's predictions
can it contribute to existing metadata as evidence o
cultural and technical history?
3. How must the archived website be described as a r
dimensional historical object to serve differing resea
access needs, while respecting the integrity of its li
evidence?
4. What are the opportunities and limitations of this ap
future historical web research and for institutional p
the KB National Library?

Current moment of progress

(aaaaah)

- Still haven't been training yet
- Technical support from the KB has fallen ill long-term:
    - no full dataset yet (almost there);
    - and GPU still not connected...
- Acquired super relevant information and ideas to move forward during the IIPC WAC conference this week
- Small start with the KG
- Re-annotation using labeling functions